ANALYTICAL FRAMEWORK FOR COMMERCIAL PRICING AND DETERMINING GLOBAL LABOR RATES

Panha Meas

AGENDA:

Task 1: Commercial Pricing Support.

- Establishes the cost floor.

Task 2: Global Labor Rate.

- Keeps it accurate year over year

SITUATION

20,000 Custom SKUs

Each with a unique cost profile. No two products cost the same

One Manufacturing Site

One site responsible for supplying the entire network

Three Regional Hub: Americas, Europe and APAC (Asia Pacific)

Each hub different freight costs, regulations and market dynamic are not the same.

TASKS

Task 1: Commercial Pricing Support

- Pillar 1: Cost to Produce
  - Direct Material Cost
  - Direct Labor
  - Scrap
  - Capstone: Forward Assembly
- Pillar 2: Discount Strategy
  - Regional
  - Loyalty
  - Performance
  - Product
  - Governance

Task 2: Global Labor Rate

- The Global Labor Rate Model
- Performance Tree
- Annual Process

Task 1: Commercial Pricing Support

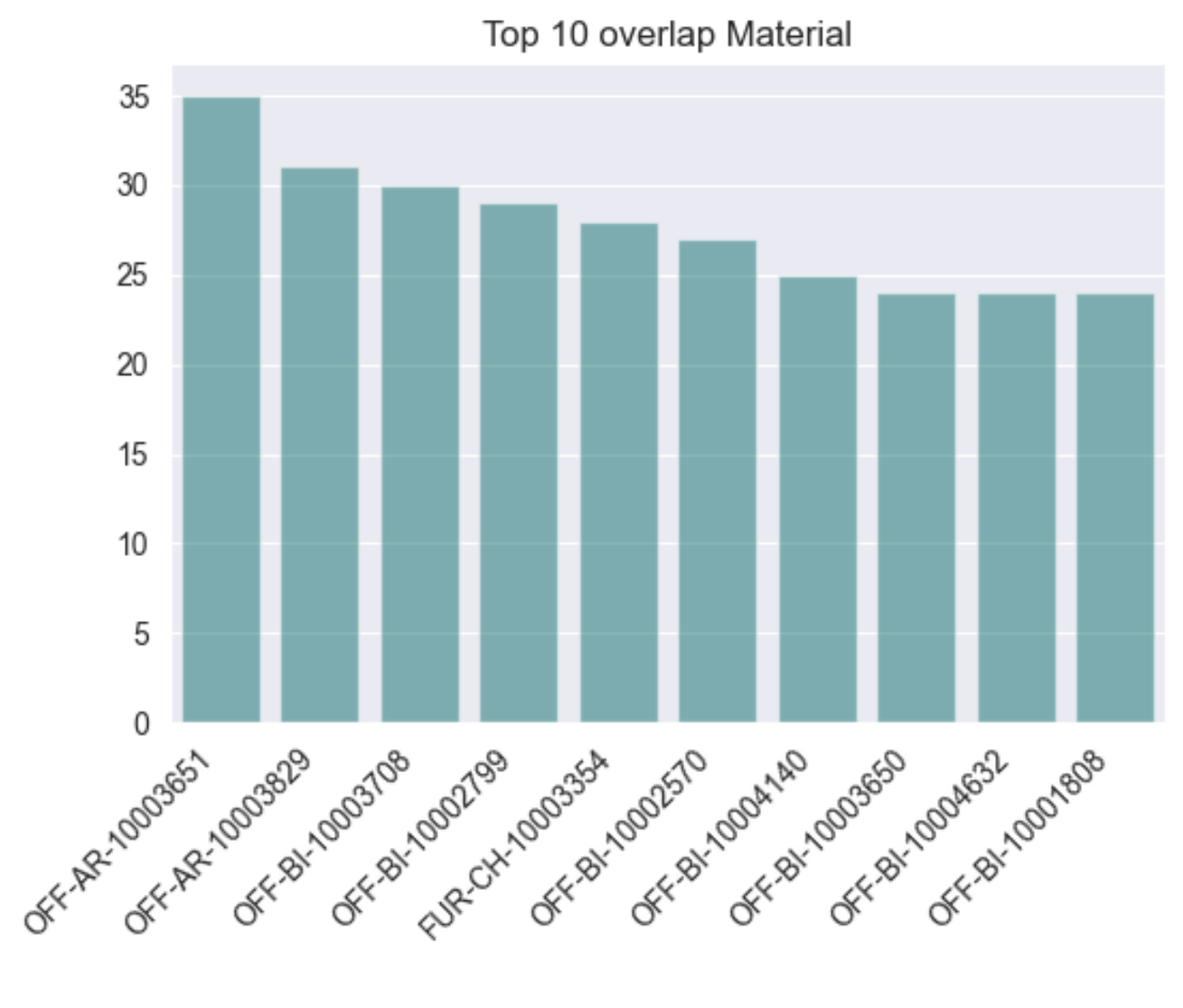
Pillar 1: Cost to Produce

You cannot offer a big discount on a product that already costs a lot to make.

Layer 1: Direct Material Cost
Supply Chain Lever: PURCHASING POWER

20,000 SKUs – unique part lists. However, one part can overlap across multiple products

Step 1: BOM (Bill of Material) Commonality
Step 2: Parts Rationalization
Step 3: Shelf-Life Optimization
Step 4: Supplier Consolidation + LTA



Layer 1: Direct Material Cost

Step 1: BOM Commonality

What: Cross-SKU parts mapping across all 20,000 SKUs to identify overlapping parts and true consolidated demand.

Why: A part used in 847 SKUs has massive negotiation leverage that nobody sees without analytics. Individual SKU buyers never see the full picture.

Data Source: ERP System (SAP) — BOM data extraction and consolidation.

Part	Category	# SKUs Using	Total Annual Consumption	Avg Unit Price	Total Annual Spend	Leverage
A11	Connector	847	125,000	$2.67	$333,750	High
A13	Filter	632	98,000	$3.27	$320,260	High
A12	Tubing	203	34,000	$1.85	$62,900	Medium
A15	Connector	45	8,000	$4.50	$36,000	Medium
A17	Bag	12	1,200	$45.00	$54,000	Low

Step 2: Parts Rationalization

What: Identify similar parts doing the same function but carrying different part numbers across the portfolio due to different designers specifying them independently.

Why: Each unique part number = separate purchase order = smaller volume = weaker negotiation. Consolidating to one preferred part multiplies purchasing power.

A11 – A11b – A11c -> Same Function

Standardize to ONE -> volume multiplies

Layer 1: Direct Material Cost

Step 3: Shelf-Life Analysis & Opportunistic Buying, Inventory Optimization

What: Determine safe maximum purchase quantities per part based on shelf-life and capture additional savings through near-expiry opportunistic buying from suppliers.
Why: Buying power increases with volume but Life Science parts have expiry dates. Must align part life with product life. Cannot buy unlimited quantity regardless of price.

Part	Monthly Demand	Shelf Life	Safe Max Buy	Remaining Life	Product Req	Alignment	Supplier Disc	Action
A11	10,417	24 mo	250,000	48 mo	39 mo	✅ Safe	30%	Buy — Priority
A13	8,167	18 mo	147,000	36 mo	39 mo	❌ Unsafe	25%	Skip
A12	2,833	6 mo	17,000	52 mo	39 mo	✅ Safe	15%	Evaluate
A17	100	36 mo	3,600	10 mo	39 mo	❌ Unsafe	40%	Skip

Layer 1: Direct Material Cost

Step 4: Supplier Consolidation + LTA (Long Term Agreement)

What: Consolidate purchasing of each part from multiple suppliers into one preferred global supplier, then lock the negotiated price with a LTA.
Why: Two separate problems solved together fragmented supplier base loses volume leverage, and even a well-negotiated price is vulnerable to repricing next year without an LTA.

LTA Decision Dataset

Part	Annual Spend	SKUs Using	Recommended LTA	Price Lock	Est. Annual Saving
A11	$333,750	847	Strategic 3yr LTA	3 years	$50,063
A13	$320,260	632	Strategic 3yr LTA	3 years	$48,039
A12	$62,900	203	Preferred 2yr	2 years	$6,290
A15	$36,000	45	Preferred 1yr	1 year	$1,800
A17	$54,000	12	Standard PO	None	$0

Layer 2: Direct Labor + Indirect Costs

Supply Chain Lever: SPEED (Flow Efficiency)

Core Idea

Labor cost is not just about how much we pay per hour. It is about how many hours it takes ONE order to travel from production start to final shipment. The gap between Touch Time and Cycle Time is where hidden labor cost lives. Speed IS our cost reduction strategy.

The Order Flow Story — Journey Of One Production Order

Step	Activity	Status
1	Raw material arrives — sits in receiving waiting for QC inspection	⏱️ Wait Point 1
2	QC inspection & documentation complete — material waiting to be staged	⏱️ Wait Point 2
3	Kitting & picking — 15 parts, 15 picks, kit travels to clean room staging	⏱️ Wait Point 3
4	Clean room entry — kit waits for assembler. Assembler waits for kit.	⏱️ Wait Point 4 — Most Painful
5	Assembly — hands on product. The ONLY value-adding step.	✅ Touch Time — 2.5 hrs
6	In-process QC check — assembly pauses waiting for inspector sign-off	⏱️ Wait Point 5
7	Assembly complete — batch record awaiting formal QA review	⏱️ Wait Point 6
8	QA batch record review — sits in queue, could be hours or days	⏱️ Wait Point 7 — Often Longest
9	Final release approved — shipped to regional hub	✅ Finish

Touch Time vs Cycle Time — The Reveal

Metric	Time	% of Total
Touch Time — actual assembly	2.5 hours	6%
Wait Time — all wait points combined	37.5 hours	94%
Total Cycle Time	**40 hours (5 days)**	**100%**

Layer 3: Non-Conformance / Scrap

Supply Chain Lever: QUALITY

The $10 Part Problem:
Real World Example $10-part damaged frequency = 5 times/day
$10 × 5 incidents/day = $50/day × 250 working days/year = $12,500/year
ON ONE PART ALONE

Across hundreds of parts in 20,000 SKUs → This number becomes enormous → Completely invisible without analytics → Quietly eating margin every single day

The Real Picture: Two Very Different Types Of Scrap

	Whole Unit Scrap	Component Level Scrap
Frequency	Rare	Every single day
Per Incident Cost	High — $450 true cost	Low — $10 per incident
Annual Impact	Moderate	High — $10,000+ per part
Visibility	✅ Always formally recorded	❌ Almost never reported
Why This Level	Inspection gates catch it early	Technician replaces quietly, no alarm

Capstone: Forward Assembly / Pre-Build Strategy

Not a single layer strategy improves ALL THREE layers simultaneously

Build once Serve many

ProductA = Common Sub-Assembly X + Unique parts for A

ProductB = Common Sub-Assembly X + Unique parts for B

ProductC = Common Sub-Assembly X + Unique parts for C

WHY build Common Sub-Assembly X three separate times?

→ Build ONCE in bulk → Store as semi-finished inventory

→ Complete final assembly when order arrives

Forecasting Requirements

Benefit	Layer	Mechanism
Lower Labor Cost	Layer 2	Build common sub-assembly once vs. multiple times
Lower Material Handling	Layer 1	Pick common parts once for all products
Lower Scrap Cost	Layer 3	Defects caught earlier at sub-assembly stage
Lower Shipping Cost	Bonus	Standard shipping replaces expedited
Higher Customer Satisfaction	Strategic	On-time or early delivery every time

PILLAR 1: CHALLENGES & RISKS

No framework is without challenges.

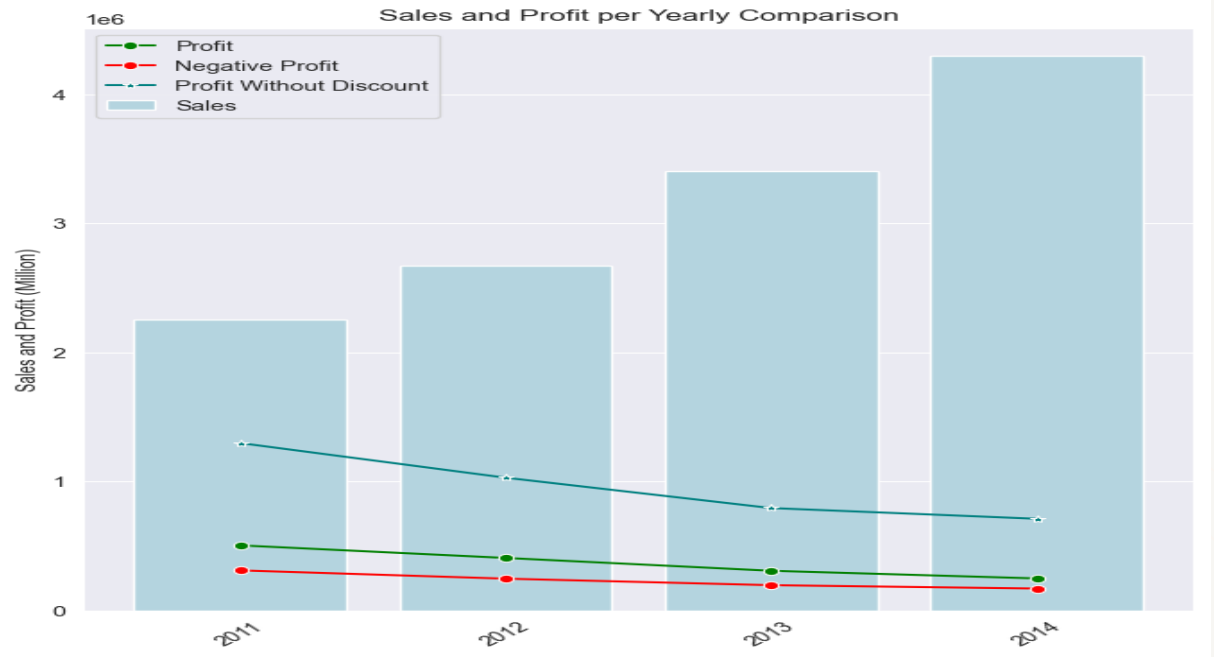
Challenges	Mitigation
BOM data quality	Data cleansing pipeline first
Shelf-life vs Bulk buying	Expiry alignment calculator
Supplier LTA commitment risk	Volume commitment cross referenced with forecast
Component scrap invisible	Digital replacement request system
Pre-build obsolescence	Demand Signal confidence score

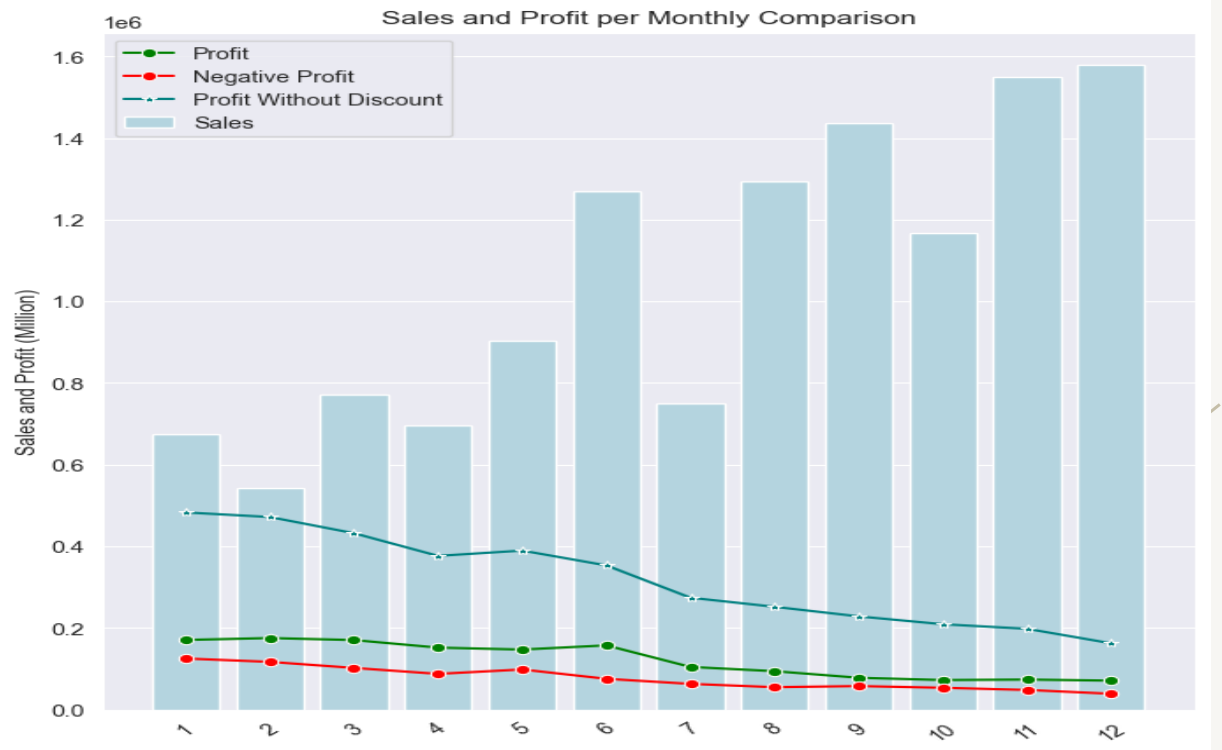
PILLAR 2: DISCOUNT STRATEGY

We do not discount to win orders. We discount to win profitable, sustainable business. Every discount must be intentional, justified, and tied to real value received in return

Sales grew. Profit declined.
Discounting was the silent killer

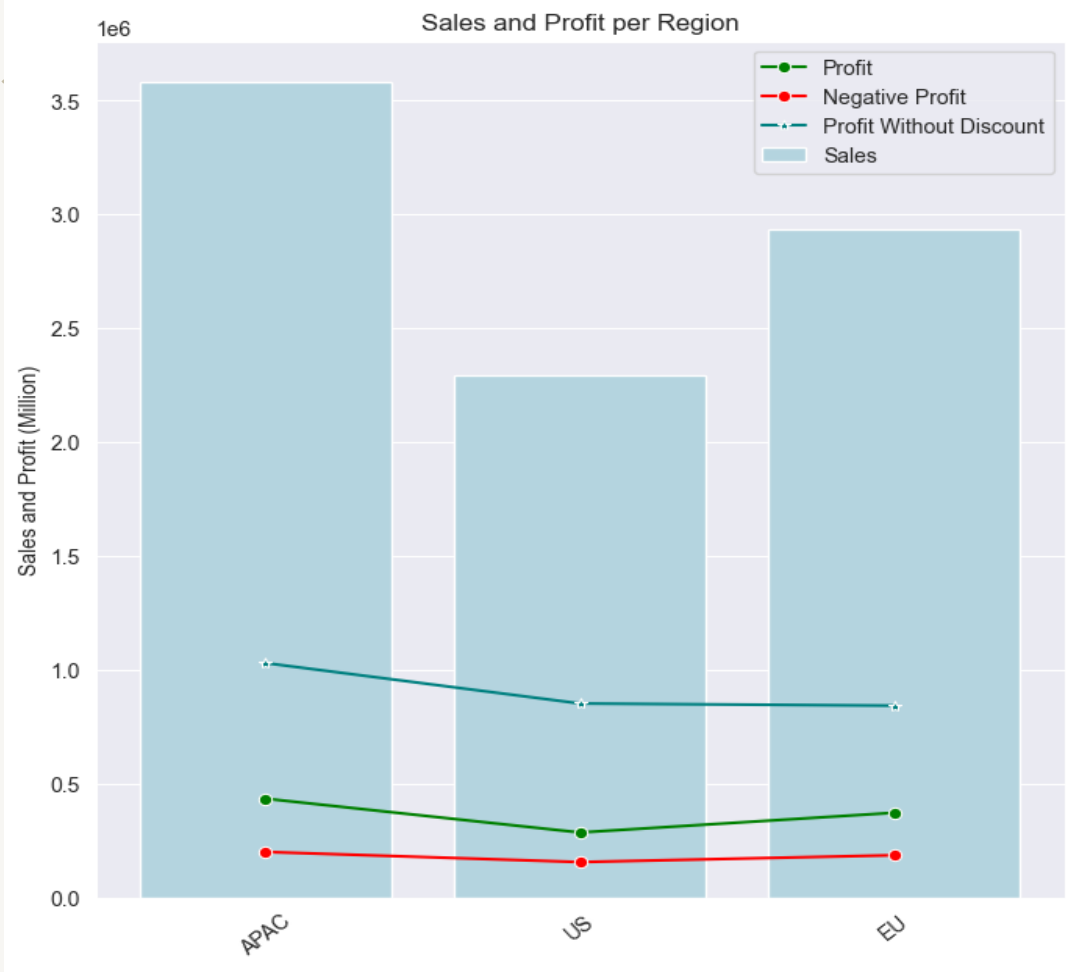
D1: Regional Floor -> Hard ceiling
D2: Loyalty -> Earn by commit
D3: Performance -> Self funded
D4: Product -> Custom vs standard
D5: Governance -> Rules engine





Dimension 1: Regional Floor – Hard Ceiling

Dimension 1 does NOT give customers a discount. It sets the ABSOLUTE WALL nobody crosses regardless of any other dimension. Discount starts at ZERO and builds through Dimensions 2, 3 and 4. The regional ceiling only activates if the earned discount exceeds what the region can sustain financially.



Regional Cost Stack — One Manufacturing Site, Three Hubs

Cost Element	Americas Hub	Europe Hub	APAC Hub
Factory Cost/Unit (same site)	$350.00	$350.00	$350.00
Outbound Freight/Unit	$12.00	$28.00	$35.00
Import Duties & Tariffs	$8.00	$22.00	$18.00
Hub Operating Cost/Unit	$15.00	$20.00	$14.00
Regulatory Compliance/Unit	$5.00	$35.00	$8.00
Inventory Holding/Unit	$8.00	$10.00	$9.00
Regional Margin Floor	**$398.00**	**$465.00**	**$434.00**
List Price	**$550.00**	**$620.00**	**$520.00**
Hard Ceiling (max possible)	**27.6%**	**25.0%**	**16.5%**

Dimension 2: Customer Loyalty Discount

Customer gives us → Predictability & volume commitment
We give them → Recognition through better pricing

Long contract = Predictable demand
Predictable demand = Better production planning
Better planning = Lower operational cost
Lower cost = We can afford to share savings back

KEY RULE: Discount earned through COMMITMENT, not history alone 5 years of buying without a contract = Standard tier History without commitment = no loyalty discount

Dimension 3: Performance Based Discount

When customer behavior reduces OUR costs
we share some of that saving back
Discount is FUNDED by savings generated
NOT taken from margin

Every performance discount must be traceable to a real operational saving we receive.

Trigger 1: Bundle Purchasing
Trigger 2: Bulk Order Discount
Trigger 3: Forecast Commitment Discount

Dimension 4: Product Based Discount

Not all products are equal

Engineering and quality investment already embedded in each SKU must be protected in pricing Customer chose custom = They pay for custom No purchasing power from unique parts No production efficiency from low volume → Minimal discount is the only responsible policy

Category	Characteristics	Max Discount	Rationale
Highly Customized SUA	Unique design, specialized components, small volume, high validation cost	0-5%	Engineering cost must be recovered — no shared purchasing power
Semi-Standard	Standard platform + modifications, moderate shared components, medium volume	5-15%	Partially amortized engineering — some purchasing power benefit
High Volume Standard	Maximum component overlap, fully amortized engineering, high volume	15-25%	Full purchasing power + production efficiency — most flexible

Dimension 5: Governance Framework

WITHOUT GOVERNANCE: Great strategy on paper Sales team discounts emotionally "I need to win this deal right now" Margins erode silently year over year Nobody notices until damage is done

Approval Authority Structure

- 0% - 3%: Sales Representative only
- 4% - 6%: Sales + Finance joint approval
- Above 6%: General Manager/Leadership

- Documentation Requirements
- No Automatic Carryover Rule
- Annual Review Checklist

Challenge:
Customer behavior cannot be predicted.
Market dynamics cannot be controlled.
Competitive pressure cannot be avoided

Full Scenario Example — Customer X, APAC

Factor	Detail	Discount Earned
Region	APAC → Hard ceiling 16.5%	—
Contract (Dimension 2)	2 year contract → Preferred tier	+7%
Bundle (Dimension 3)	Regular A+B+C bundle orders	+3%
Forecast (Dimension 3)	6-month rolling forecast provided	+1%
Product (Dimension 4)	High volume standard product	+8%
Total Earned		**19%**
Regional Ceiling	APAC hard ceiling applies	**16.5%**
Final Approved Discount	Ceiling overrides earned	**16.5%** ✅

Task 2: Global Labor Rate

A cost floor is only valuable if it stays accurate year over year.

Situation:	Task:
One site generate savings annually across purchasing, labor and quality but without a structured process those savings never reach the cost baseline	Design an annual process to: • Consolidate savings across the network • Calculate a fair global labor rate • Feed it into next year's pricing decisions

Without structure:
Savings lost – Rate stale
Wrong foundation

With structure:
Savings captured – Rate current
Every bid backed by real data

Block 1: The Global Rate Model

A performance-based percentage rate applied to local currency base rates across all three regional hubs.

Americas – Europe – APAC
Same % - different currency
Same logic – different reality

Global performance sets the BAND

Exceptional -> 5% to 12%
Strong -> 4% to 10%
On Track -> 3% to 7%
Below Target -> 2% to 5%
Weak -> 1% to 3%

CRITICAL PRINCIPLE:

Rate must always live within what the business actually earned
If company grew profit by 5%
giving employees 8% rate increase means
labor cost growing faster than revenue — unsustainable

Block 2: Performance Tree

Band minimum -> everyone starts here

Gate 1: Regional -> 37.5% of band
Gate 2: Team -> 25.0% of band
Gate 3: Individual -> 37.5% of band

FIVE PERFORMANCE CATEGORIES PER GATE:

Exceptional (100%) → Significantly exceeded all targets
Above Average (75%) → Exceeded most targets consistently
On Target (50%) → Met expectations reliably
Below Target (25%) → Partially met — improvement needed
Unsatisfactory (0%) → Did not meet targets

The 10 Point Scoring System

Gate	Weight	Max Points	Exceptional (100%)	Above Avg (75%)	On Target (50%)	Below Target (25%)	Unsat (0%)
Regional	37.5%	3.75	3.75	2.81	1.88	0.94	0.00
Team	25.0%	2.50	2.50	1.88	1.25	0.63	0.00
Individual	37.5%	3.75	3.75	2.81	1.88	0.94	0.00
Total	**100%**	**10.00**	**10.00**	**7.50**	**5.00**	**2.50**	**0.00**

The Rate Calculation Formula

```
FINAL RATE =
  Band Minimum
+ (Total Score ÷ 10) × Band Width

WHERE:
Band Minimum = Set by global performance tier
Band Width   = Maximum − Minimum
Total Score  = Sum of all three gate scores out of 10

EXAMPLE — Strong year (Band: 4% to 10%):
Regional    = Exceptional  = 3.75 pts
Team        = On Target    = 1.25 pts
Individual  = Exceptional  = 3.75 pts
Total Score = 8.75 / 10   = 87.5%

Final Rate  = 4% + (87.5% × 6%) = 4% + 5.25% = 9.25%
```

Full Scenario Matrix

Global Band	Regional	Team	Individual	Score	Final Rate	Story
2-10% (Strong)	Exceptional	Exceptional	Exceptional	10.00	10.0%	Everything firing — maximum reward
2-10% (Strong)	Exceptional	On Target	Exceptional	8.75	9.0%	Team average — still strong result
2-10% (Strong)	On Target	On Target	On Target	5.00	6.0%	Solid average year across board
2-10% (Strong)	Unsatisfactory	Unsatisfactory	Unsatisfactory	0.00	2.0%	Minimum floor protects everyone
1-3% (Weak)	Exceptional	Exceptional	Exceptional	10.00	3.0%	Strong performance — ceiling caps

Block 3: Annual Process

A structured, repeatable governance cycle that runs every year from locking the books to monitoring the output.

Step 1: Define The Performance Period & Lock The Books

Step 2: Set the global band

Step 3: Validate regional savings

Step 4: Run the performance tree

Step 5: Review, approve & communicate

Step 6: Update the cost baseline

Step 7: Monitor, audit & improve

A global labor rate that is:

FINANCIALLY DISCIPLINED:
Never exceeds what the business can afford
Global band ceiling enforced by data

PERFORMANCE DRIVEN:
Clear line of sight for every employee
Regional + Team + Individual contribution recognized

REGIONALLY FAIR:
Local currency and inflation considered
Same process — different regional outcomes

COMMERCIALLY CONNECTED:
New rate feeds directly into Task 1
Margin floors updated
Discount ceilings recalibrated

CONTINUOUSLY IMPROVING:
Each year the process gets smarter
Savings compound over time
System never goes stale

TASK 2: CHALLENGES & RISKS

Challenge	Mitigation
Savings overclaimed	System validation vs manager report
Political band pressure	Anchor to certified data
Talent retention weak year	Non-monetary recognition
Inflation exceeds band minimum	HR & Leadership special review
Rate update too late	Hard calendar deadline locked

THANK YOU

Panha Meas